Detailed Summary of the v7 ETH Report

Four-State Boolean Reduction, Closed Higher-Mode Summation, Exactness, Speedups, Memory, and Full-Algorithm Limits

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Abstract

This report gives a self-contained summary of revision 7 of the Ram Calculus investigation of the manuscript *The Exponential Time Hypothesis is False*. V7 implements and benchmarks two exact identities already established in v6: a six-coordinate packet polynomial is reduced to four Boolean coordinates, and an explicit sum of nonzero higher geometric modes is replaced by a finite closed expression. On four two-edge-packet fixtures, all 324 worker outputs reproduce independent exact counts, and all 288 non-direct outputs reproduce the complete root local polynomial. With the conservative source precision schedule, the combined reduction gives ratios of baseline to revised median elapsed time of 5.24–6.46; with v6's degree-aware precision and certified symmetry, the corresponding ratios are 1.52–2.07. These are local, host-dependent measurements, not full SAT speedups. Separate allocation profiling finds less than one percent change in combined-method peak tracked allocations, with some increases. We explain the quotient proof, the closed-sum proof, protected division, orbit and cache effects, exactness controls, timing variability, and failure cases. V7 adds implementation and validation evidence, not a new precision theorem, improved numerical accuracy, deeper-recursion experiment, or general ETH complexity result. The methods are also available through conventional Boolean algebra and finite-ring arithmetic; no Ram-exclusive computational advantage is established. This summary uses the saved v7 report and experimental records without rerunning benchmarks or re-evaluating the source manuscript's full claim.

Contents

1 Main conclusions and what is new in v7 3

2 Matched instances, variants, and computational setting 3

3 The exact four-state Boolean reduction 4

4 Closed-form summation of the nonzero higher modes 6

5 Symmetry, child interfaces, and unchanged precision 8

6 Test protocol and exactness evidence 9

7 Measured runtime improvements and their variability 10

8 Operation counts: why the implementation is faster 12

9 Memory results: a better measurement, not a large saving 12

10 Failure cases and boundaries of validity 14

11 Role of Ram Calculus and comparison with conventional methods 14

12 What remains necessary for the full algorithm 15

13 Evidence provenance and conclusion 16

1 Main conclusions and what is new in v7

The scientific contribution of v7 [1] is best separated into three layers: exact algebra, a tested implementation, and limits on what the implementation establishes.

1. **Two already-proved identities become executable.** V6 [2] identified the four-state quotient and a closed formula for the coalesced positive-order geometric coefficient. V7 pays their actual implementation costs and compares them against the frozen v6 routine. The identities are not newly discovered theorems in v7.
2. **Both reductions preserve more than the final answer.** They preserve the packet kernel at every relevant Boolean observation. The measured implementations also reproduce every coefficient of the root local polynomial, including all extension-ring coordinates. Agreement of a single final count is therefore not the only check.
3. **The combined reductions are faster in the median on all four fixtures under both precision schedules.** The larger gains occur when the original conservative precision is retained, because the explicit geometric loop is longer at that precision.
4. **No substantial peak-memory reduction is measured.** The improved allocation experiment resolves transient Python allocations, but the combined variants change the recorded peaks by less than one percent. Some peaks increase.
5. **No improvement to the full source algorithm is proved.** The tests involve four edge variables, two packets, and one active recursive level. They neither implement the full SAT pipeline nor establish a new worst-case time or space bound.

The new v7 section occupies physical pages 6–14 of the 89-page report. The retained v6 scientific body starts on physical page 15. Historical statements in that retained material that a reduction was unbenchmarked are superseded only for the packet experiments reported in v7. The earlier deeper-recursion results remain historical evidence; they are not new v7 tests.

Meaning of "original routine." Here this means the frozen *finite, rescaled source-form contraction and Hermite-recovery routine implemented in v6*. It is not an executable full ETH/SAT implementation provided by the OpenAI manuscript's authors. Its conservative-precision branch is one comparator; its existing degree-aware and symmetry-reduced branch is a second comparator. Keeping that distinction prevents a local speed ratio from being misreported as an improvement to the manuscript's complete algorithm.

2 Matched instances, variants, and computational setting

2.1 What the four fixtures contain

Every timed root has four Boolean edge variables, arranged in two packets. Each packet has two real positions, 0 and 2, and genuinely coupled two-port tensors. The root nevertheless decomposes across the two packets. Its four generated two-edge children are evaluated as leaves at threshold two. Thus there is one active interpolation and contraction level, followed by direct leaves.

All real interactions in this setup are crossing interactions. General mixtures of local and crossing edges are not exercised. In particular, the v6 three-active-level singleton family, D01, is neither

modified nor rebenchmarked by v7. No v7 experiment combines deeper recursion with general multi-edge packets.

The fixture definitions and exact counts are as follows. Tables at a Boolean two-port vertex are ordered as four entries. P01 uses OR tables $(0, 1, 1, 1)$ at all four vertices. P02 replaces the first row vertex by XOR, $(0, 1, 1, 0)$. P03 replaces it by the zero table. P04 uses integer weight $2^{15} + 7i + 3j + 1$ at entry j of vertex i.

Case	Exact answer	Output bits	Direct median (μs)
P01	9	5	228.1
P02	6	5	228.9
P03	0	5	201.3
P04	18482797547933526180	65	244.8

P04 is a weighted contraction, not a Boolean SAT count. Its value exceeds 2^{64}. Proven nonnegative upper bounds for the selected fixtures are below the respective output moduli, so matching residues identify the exact integers, rather than merely congruence classes. Direct enumeration has only $2^4 = 16$ assignments and is overwhelmingly faster than all interpolation-based variants on these tiny instances.

2.2 Nine variants and two distinct comparisons

The suite uses direct enumeration and eight recursive variants:

Symbol	Meaning	What remains fixed
S	Conservative v6 source precision	No packet symmetry reduction
DG	Degree-aware precision and certified symmetry	Existing v6 schedule
6 / 4	Six / four observation coordinates per packet	Same two real edges
L / C	Explicit geometric loop / closed sum	Same nonzero higher modes

For example, S4C combines the four-state basis and the closed sum under source precision; DG6L is the existing degree-aware/symmetry method before either new implementation change. The eight names are S6L, S4L, S6C, S4C, DG6L, DG4L, DG6C, and DG4C.

The 6L comparators execute the frozen v6 implementation, not a newly rewritten approximation of it. A 4L variant changes only the observation basis; a 6C variant changes only higher-mode coefficient construction; a 4C variant changes both. These matched ablations permit a more useful interpretation than comparing only the fastest new variant with the slowest old variant.

The same creation-time lifts, child construction, exact-key 16-entry cache, interpolation plan, and 116 local evaluations are retained. Crucially, the four-state routine first builds the original six coefficients and then folds them. Its benchmark includes that folding cost. Directly constructing only four coefficients remains a possible optimization, not a measured v7 result.

3 The exact four-state Boolean reduction

3.1 Why six coordinates contain redundancy

The v6 row kernel is represented by a polynomial

$$P(Z_0, Z_+) = c_{00} + c_{10}Z_0 + c_{01}Z_+ + c_{20}Z_0^2 + c_{11}Z_0Z_+ + c_{02}Z_+^2. \tag{1}$$

Its arguments are not arbitrary independent real numbers. For two Boolean column bits x, y, the actual observations are $Z_0 = x + y$ and $Z_+ = y$. Consequently the kernel is a function on only four column states. The identities $x^2 = x$ and $y^2 = y$ allow a smaller exact coordinate representation.

Proposition 3.1 (Exact Boolean quotient)**.** *For every commutative coefficient ring R, including rings with zero divisors, the restriction of (1) to these observations is*

$$P(x + y, y) = d_{00} + d_{10}x + d_{01}y + d_{11}xy, \tag{2}$$

where

$$\begin{aligned} d_{00} &= c_{00}, & d_{10} &= c_{10} + c_{20}, \\ d_{01} &= c_{10} + c_{01} + c_{20} + c_{11} + c_{02}, & d_{11} &= 2c_{20} + c_{11}. \end{aligned} \tag{3}$$

No division, field-rank assumption, or approximation is required.

Proof. Work in $R[x, y]/(x^2 - x, y^2 - y)$. Then

$$(x + y)^2 = x + y + 2xy, \qquad (x + y)y = xy + y, \qquad y^2 = y.$$

Substitution into (1) and coefficient collection give (3). Moreover, evaluating the ordered basis $(1, x, y, xy)$ at $(0, 0), (1, 0), (0, 1), (1, 1)$ gives

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix},$$

whose determinant is one. Evaluation is invertible over any such ring. Equivalently, for function values $f_{00}, f_{10}, f_{01}, f_{11}$, the coefficients are f_{00}, $f_{10} - f_{00}$, $f_{01} - f_{00}$, and $f_{11} - f_{10} - f_{01} + f_{00}$. The quotient is therefore exact even when ordinary division by a nonunit would be invalid. □

The proof establishes equality for each fixed row-bit and column-bit kernel value, not merely after summing an entire network. Therefore contraction with the unchanged row and column tensors preserves the result at each interpolation point. This is the algebraic reason that the local polynomial and final integer remain the same.

3.2 What must change in the implementation

The column-side functions must change along with the row coefficients. The old basis uses monomials $Z_0^a Z_+^b$; the new one uses $x^a y^b$ for $(a, b) \in \{0, 1\}^2$. Pairing the folded coefficients with the old column basis would not implement the identity. Four states refer to observation coordinates, not to fewer Boolean edges or a shallower recursive network.

For two packets the observation assignments fall from $6^2 = 36$ to $4^2 = 16$. The four Boolean basis functions span all functions on a two-bit packet, but this does not produce a four-dimensional basis for an arbitrary larger packet. A generic k-bit packet has 2^k Boolean function coordinates. Any further compression needs additional algebraic structure or a proved restricted function space.

3.3 Executed controls that expose incorrect simplifications

Take $P = Z_0^2$. Its correct quotient is $x+y+2xy$. At $(1,1)$ modulo eight the answer is four. Omitting the entire $2xy$ contribution gives two. Merely using characteristic-two intuition for a coefficient ring modulo 2^B is therefore incorrect: its residue field has characteristic two, but the working ring generally does not.

Nor can the Boolean quotient be applied off its domain. At $(x, y) = (2, 3)$ modulo 16, the original square is $(x + y)^2 = 9$, whereas the reduced expression is $x + y + 2xy = 1$. The exactness claim is strong on the specified Boolean domain and false as an unrestricted polynomial identity.

4 Closed-form summation of the nonzero higher modes

4.1 Ring, kernel, and coalesced observations

V7 retains the v6 coefficient ring and interpolation centers:

$$R_B = (\mathbb{Z}/2^B\mathbb{Z})[U]/(U^5 + U^2 + 1), \qquad T = t + 8\zeta, \qquad t \in \{-3, -1, 1, 3\}.$$

Elements have five coordinates. The symbols B and ζ denote working precision and an extension-ring evaluation point, respectively. Define

$$\begin{aligned} C(T) &= \prod_{s=-4}^{3} (T - s), & C'(0) &= -144, \\ N &= C(T)/(-144), & A_x &= N(T + x)^{-1}, & r_x &= 2(T + x)^{-1}, \end{aligned} \tag{4}$$

for row position $x \in \{0, 2\}$. Both $T + x$ and $T + x - 2$ are units. The normalized numerator N is computed with protected division as explained below; the notation does not mean that 144 has an inverse in the working ring.

The two column positions are $2j$ for $j = 0, 1$. Writing their Boolean bits as X_0, X_2, the observed moments satisfy

$$L_0 = X_0 + X_2, \qquad L_b = \sum_{j=0}^{1} j^b X_{2j} = X_2 \quad (b \geq 1).$$

Thus all positive-order moments are the same function, although their coefficients are generally nonzero. V6 already coalesces their observation into one coordinate, but constructs its coefficient through the explicit finite loop

$$B_x = \sum_{b=1}^{B-1} A_x r_x^b. \tag{5}$$

V7 replaces this coefficient loop, not the underlying higher-mode contribution, by a closed expression.

Proposition 4.1 (Exact finite closed sum)**.** *In R_B,*

$$B_x = N\left((T + x - 2)^{-1} - (T + x)^{-1}\right). \tag{6}$$

This gives exactly the same packet kernel coefficient as (5) modulo 2^B.

Proof. Since r_x is divisible by two, $r_x^B = 0$. Multiplication gives

$$(1 - r_x) \sum_{b=0}^{B-1} r_x^b = 1 - r_x^B = 1.$$

Also $1 - r_x = (T + x - 2)(T + x)^{-1}$ is a unit. Multiplying the finite geometric identity by A_x and subtracting its order-zero term gives

$$\sum_{b=1}^{B-1} A_x r_x^b = \frac{N}{T + x} \left(\frac{T + x}{T + x - 2} - 1 \right),$$

which is (6). The fractions here involve only the stated units. No analytic convergence assertion or floating-point approximation is used. □

4.2 Why the local arithmetic becomes cheaper

At fixed extension degree, the old loop can require a number of additional modular products and additions linear in B. The closed coefficient needs a constant number of products and differences beyond the two unit inversions. Those inversions are not an extra hidden cost of the new method: the frozen v6 timed loop already computes both inverses and the second closed kernel value, even with diagnostics off. Thus the comparison really removes the loop's arithmetic, rather than removing a validation task charged only to the baseline.

The implemented Newton inversions use a number of lifting stages logarithmic in B. Arithmetic within each stage still has a bit cost depending on precision. Consequently "constant number of additional ring operations" must not be confused with constant bit complexity, or with a bound for the complete network contraction.

In a diagnostic worker the old loop separately checks the closed result. That extra check has its own counters and is outside all timing samples. A zero geometric-loop counter in a closed variant means that the loop was eliminated, *not* that the higher-order terms were omitted.

4.3 Protected division and a concrete negative control

The factor $-144 = 16(-9)$ has a nonunit power of two. To recover N modulo 2^B, the implementation first computes numerator coordinates modulo 2^{B+4}, divides their canonical representatives by 16, then multiplies by the inverse of -9 modulo 2^B.

The reason for the four guard bits is elementary. If an exact numerator coordinate c is divisible by 16, knowledge of c modulo 2^{B+4} determines $c/16$ modulo 2^B. Knowledge only modulo 2^B generally does not. At the permitted centers the numerator is actually divisible by 2^7, but this does not justify erasing the precision needed for the specified division.

The executed scalar control uses $T = 5$ and $B = 5$. Here $C(5) = 362880$ and $C(5)/(-144) = -2520 = 8 \pmod{32}$. Reducing the numerator to zero modulo 32 before dividing produces zero instead of eight. Closed summation leaves the four guard bits unchanged.

4.4 Worked example combining both reductions

At $T = 5$, the two row positions give

$$(A_0, B_0) = (-504, -336), \qquad (A_2, B_2) = (-360, -144).$$

For row bits $(0,0)$ the packet polynomial is

$$P(Z_0, Z_+) = (1 + 504Z_0 + 336Z_+)(1 + 360Z_0 + 144Z_+).$$

Its six coefficients are

c_{00}	c_{10}	c_{01}	c_{20}	c_{11}	c_{02}
1	864	480	181440	193536	48384

and the Boolean quotient becomes

$$P(x + y, y) = 1 + 182304x + 424704y + 556416xy. \tag{7}$$

All equalities may be read as exact integer equalities or modulo an appropriately chosen power of two. At the four Boolean points, (7) gives

(x, y)	$(0,0)$	$(1,0)$	$(0,1)$	$(1,1)$
Value	1	182305	424705	1163425

At $(0,1)$, discarding all positive-order contributions would instead give $(1 + 504)(1 + 360) = 182305$, which is wrong. Both v7 reductions retain the correct value 424705. This example shows explicitly that a smaller coordinate system and a closed sum can preserve information that an unjustified truncation destroys.

5 Symmetry, child interfaces, and unchanged precision

5.1 What the orbit reduction actually certifies

For P01, interchange of the two whole packets is certified. The number of unordered observation representatives is then $6(6 + 1)/2 = 21$ in the old basis and $4(4 + 1)/2 = 10$ in the new basis. P02–P04 do not admit that interchange certificate and retain 36 or 16 representatives, respectively.

The certificate preserves ordered real positions, tensor values and axes, row/column side, creation precision, and the grouping up to a bijection. It does not grant arbitrary swaps of coordinates inside a packet. Orbit contributions are multiplied by integer multiplicities; they are never averaged by dividing by the group order. This matters because two need not be invertible in the coefficient ring.

Generated row and column tensors continue to have four entries indexed by their two Boolean ports. Their creation lifts, child precision requests, construction, and exact-key cache policy are unchanged. A different basis nevertheless changes the keys and values encountered, so the number of executed leaves need not fall in exactly the same ratio as the number of observation assignments.

5.2 What v7 does not change about precision

For these fixtures the deformation uses polynomial kernels of degree seven, with a root degree bound $7M = 28$ at $M = 4$. The retained recovery procedure uses 29 points at each of four centers, hence 116 local evaluations. Both reductions preserve the same deformation polynomial; they neither lower this degree bound nor shorten the recovery plan.

Fixtures	Output bits	S working	S numerator	DG working	DG numerator
P01–P03	5	485	489	54	58
P04	65	545	549	114	118

The numerator columns include the four protected-division guard bits. These values are identical within a schedule for 6L, 4L, 6C, and 4C. The reduction from source to DG precision is inherited from v6's degree-aware reconstruction and valuation bounds, not a new v7 result. The v6 deep-family precision path also remains unchanged and was not executed in this suite.

This explains an important interaction: once DG has already shortened the explicit loop through lower certified working precision, replacing that loop by a closed expression has less work left to remove. The different speed ratios under S and DG are therefore coherent with the operation counts, without implying a new precision saving in v7.

6 Test protocol and exactness evidence

6.1 Worker counts and independent oracles

Four fixtures and nine variants give 36 case–variant combinations. Each has five untraced timing workers, three separate memory-profile workers, and one diagnostic worker:

$$36 \cdot 5 = 180, \qquad 36 \cdot 3 = 108, \qquad 36 \cdot 1 = 36, \qquad \text{total} = 324.$$

All 324 outputs match the independent exact final count. The eight non-direct variants have 288 outputs in total; every one matches the complete root local polynomial. For each fixture this amounts to 81 count matches and 72 full-polynomial matches.

The polynomial check covers 29 coefficients, four centers, and all five extension coordinates at the exact precision required by each plan. The independent reference construction uses exact rational polynomial arithmetic, rather than invoking the new kernel engine. This reduces the risk that an implementation error is accepted simply because the same error was reproduced in the oracle.

6.2 Additional controls and record auditing

The saved evidence [3, 4, 5] includes:

- Eight no-op fork comparisons, one for each fixture and S/DG schedule, that match all non-time fields, including counters, precision plans, node records, and root polynomials.
- 192 closed-versus-loop row-kernel checks at six precisions, including non-base extension-ring evaluation points.
- 1536 fixed row/column-bit kernel-product comparisons after Boolean folding, plus 192 arbitrary-coefficient ring evaluations.
- Twelve additional deterministic weighted fixtures generated with seed 722007, checked with four DG variants, giving 48 additional full-polynomial validations. They are correctness controls, not a timed random-instance performance study.
- An independent post-audit that rejects 398 deliberately mutated records or unsupported-domain requests. It checks the complete worker manifest, expected precision, plans and weights, coordinate shape and canonicality, variant identity, and counter relations.

The supported public experimental domain is explicitly the four-edge packet fixture with leaf threshold two. Lower-level functions are not claimed to constitute a validated arbitrary-network

API. Cache conservation and arithmetic relations are audited, but the post-audit does not replay the entire cache execution history.

These checks support correctness for the implemented identities and tested finite scope. They do not replace a proof that the full source algorithm's recursion, assumptions, and complexity analysis are valid.

6.3 Timing boundaries and environment

Workers run sequentially in fresh processes with rotated method order. Timing excludes interpreter startup, imports, independent oracle construction, and extra diagnostic comparisons. It includes network construction, precision and symmetry planning, protected arithmetic, coefficient folding, child construction and cache use, interpolation, and target recovery. CPU time is recorded alongside elapsed time.

The environment recorded in the suite is CPython 3.12.14 on Windows 11 build 26300, with an Intel64 Family 6 Model 140 Stepping 1 processor. Five timing samples per combination are sufficient for a descriptive comparison on this host, not a broad statistical performance claim. Memory profiling is performed separately; its slower traced timings are never mixed into the timing medians.

7 Measured runtime improvements and their variability

7.1 All ablation medians

The following two tables report elapsed seconds, using medians of the five untraced timing workers. "Combined ratio" means the 6L median divided by the 4C median within the same fixture and schedule.

Case	S6L	S4L	S6C	S4C	Combined ratio
P01	3.0953	2.6546	0.8731	0.5908	5.24×
P02	3.1716	2.6410	0.7737	0.5204	6.10×
P03	2.6004	2.4440	0.6097	0.4544	5.72×
P04	3.9597	4.1678	0.8958	0.6126	6.46×

Case	DG6L	DG4L	DG6C	DG4C	Combined ratio
P01	0.5333	0.4415	0.4158	0.3508	1.52×
P02	0.6242	0.5234	0.5642	0.4088	1.53×
P03	0.5882	0.4414	0.4824	0.3465	1.70×
P04	0.8534	0.6548	0.6055	0.4120	2.07×

Separating the reductions gives the following ratios. Each numerator is the 6L median; a ratio below one is a median slowdown.

	S ratios			DG ratios		
Case	/4L	/6C	/4C	/4L	/6C	/4C
P01	1.17	3.55	5.24	1.21	1.28	1.52
P02	1.20	4.10	6.10	1.19	1.11	1.53
P03	1.06	4.26	5.72	1.33	1.22	1.70
P04	0.95	4.42	6.46	1.30	1.41	2.07

Closed summation is the dominant measured change under source precision. Under DG, both reductions provide more modest additional benefits. Their effects should not be assumed independent or multiplied from unrelated experiments: the basis, loop, precision, symmetry, and cache all interact.

7.2 The important weighted-case failure

On P04 under S, folding alone has median 4.1678 seconds against 3.9597 seconds for S6L, about 5.3 percent slower. It nevertheless executes fewer logical observation assignments. This is a direct counterexample to assuming that a smaller basis must always yield a smaller measured elapsed time.

The five-run ranges overlap substantially: S6L spans 3.840–6.128 seconds, while S4L spans 3.299–5.294 seconds. Thus the evidence records a descriptive median slowdown; it does not isolate an inherent cause or establish statistical significance. Extra folding work, changed cache behavior, and host scheduling are possible influences, but their individual causal contributions were not experimentally separated.

7.3 Timing spread and CPU confirmation

The table shows the elapsed min–max ranges for the combined comparison, along with ratios formed by pairing the same repeat index. Such pairs are still sequential measurements, not simultaneous controlled trials.

Case	Schedule	6L seconds	4C seconds	Paired ratios: min / med / max
P01	S	2.709–6.459	0.517–0.706	4.45 / 5.38 / 9.14
P02	S	2.688–3.731	0.496–0.588	5.42 / 5.95 / 7.17
P03	S	2.535–2.784	0.441–1.024	2.48 / 5.77 / 5.90
P04	S	3.840–6.128	0.537–0.623	6.16 / 6.35 / 11.42
P01	DG	0.493–0.670	0.311–0.746	0.90 / 1.52 / 1.83
P02	DG	0.601–0.779	0.385–0.502	1.52 / 1.56 / 1.65
P03	DG	0.561–0.690	0.321–0.366	1.53 / 1.68 / 2.15
P04	DG	0.833–0.885	0.402–0.457	1.88 / 2.07 / 2.15

The minimum P01/DG paired ratio is 0.90: even the combined method is not faster in every individual repeat. A median of paired ratios is also not the same statistic as a ratio of medians, explaining small apparent differences between the tables.

The CPU medians support a genuine arithmetic-work reduction, while not eliminating all experimental limitations:

Case	S6L CPU seconds	S4C	DG6L	DG4C
P01	3.0781	0.5781	0.5312	0.3438
P02	3.1562	0.5312	0.6250	0.4062
P03	2.6094	0.4531	0.5938	0.3438
P04	3.9219	0.6094	0.8438	0.4062

The reported speedups remain descriptive, fixture-specific measurements. They are not confidence intervals, asymptotic estimates, improvements over the best SAT solver, or performance guarantees on all inputs.

8 Operation counts: why the implementation is faster

The diagnostic counters distinguish eliminated work from inferred complexity. “Reps” below is the representative count per local evaluation; requests and the remaining counts cover the root run. Child requests include cache hits, while leaf calls count executed leaves. Ring products also include arithmetic outside kernel construction, such as guarded arithmetic and interpolation.

P01 variant	Reps	Child requests	Leaf calls	Loop orders	Ring products
S6L	36	16704	570	107976	162358
S4L	16	7424	345	107976	151278
S6C	36	16704	570	0	54382
S4C	16	7424	345	0	43302
DG6L	21	9744	570	11432	55896
DG4L	10	4640	345	11432	48992
DG6C	21	9744	570	0	44464
DG4C	10	4640	345	0	37560

For example, S6L to S6C removes exactly 107976 counted ring products, matching the eliminated loop orders. S4L has the same loop-order count as S6L because it changes the observation basis, not the row coefficient construction. DG starts with a much shorter loop because its working precision has already been reduced.

The weighted fixture has different cache reuse and no packet-interchange certificate:

P04 variant	Reps	Child requests	Leaf calls	Loop orders	Ring products
S6L	36	16704	1819	121416	186776
S4L	16	7424	913	121416	170248
S6C	36	16704	1819	0	65360
S4C	16	7424	913	0	48832
DG6L	36	16704	1819	24872	87274
DG4L	16	7424	913	24872	70746
DG6C	36	16704	1819	0	62402
DG4C	16	7424	913	0	45874

For every four-state run there are 464 coefficient folds, from four row-bit assignments at each of 116 evaluations. Each closed-sum run uses 232 closed row calculations, two per evaluation. Logical observation terms fall from $36 \cdot 116 = 4176$ to $16 \cdot 116 = 1856$ before symmetry. The 116 interpolation evaluations themselves are unchanged.

These counters provide an explanation stronger than timing alone: the code actually eliminates the intended arithmetic and observation work. They still do not imply a runtime ratio of exactly 36/16, because interpolation, planning, arithmetic precision, folding, and cache effects do not all scale with that count.

9 Memory results: a better measurement, not a large saving

9.1 What is measured

V7 supplements the process high-water mark with separate `tracemalloc` runs. Tracing starts after imports and fixture selection but before the pipeline; garbage collection precedes each profile. The

principal quantity is the increase of peak tracked Python allocation above the recorded pre-call baseline, in KiB (1024 bytes). It includes tracked allocations for the pipeline, cache, interpolation, and returned result. It does not include the whole interpreter or preexisting input/module state and does not capture every native allocation.

A transient-allocation self-test allocates and releases a 2 MiB `bytearray` before returning. It records baseline zero, current 160 bytes, and peak 2097369 bytes. This shows that the metric can see a freed transient peak that a return-state measurement would miss. It does not establish that all memory used by arbitrary native code is covered.

9.2 All allocation medians

Each entry is the median of three separate memory workers. The last column is $100(4C/6L - 1)$, so positive means an increase.

Case	S6L KiB	S4L	S6C	S4C	Combined change (%)
P01	818.82	819.26	817.56	819.20	+0.05
P02	809.08	807.85	807.82	807.78	−0.16
P03	783.57	786.61	783.51	786.54	+0.38
P04	835.19	828.46	833.85	828.40	−0.81

Case	DG6L KiB	DG4L	DG6C	DG4C	Combined change (%)
P01	537.80	537.38	537.16	537.35	−0.08
P02	534.86	532.72	534.23	532.69	−0.41
P03	526.78	525.97	526.75	525.94	−0.16
P04	568.49	563.64	567.77	563.61	−0.86

Every combined change has magnitude below one percent. P01/S and P03/S increase. For the displayed 6L/4C pairs, all three raw peak observations are identical within a case and variant, not merely equal after rounding. This is repeatability for these fixed Python runs, not a universal memory guarantee.

Reducing the observation basis does not necessarily reduce the maximum live allocation. The implementation still constructs the old six coefficients, and interpolation storage, child objects, cache state, and returned values remain. These are plausible contributors, but the suite does not contain an allocation-by-stage attribution. A numerical state-count ratio such as 6/4 is not a measured peak-memory ratio.

DG has markedly smaller tracked peaks than S, but that comparison includes v6's prior precision and symmetry changes. It must not be credited entirely to the new v7 reductions.

9.3 Why process RSS still cannot prove a space improvement

Untraced timing-worker process peaks range from approximately 25.65 to 26.34 MiB. In 161 of the 180 timing workers the high-water increase above the pre-call process peak is zero. Interpreter, imports, and earlier allocations therefore obscure the algorithm-specific peak in that metric.

The defensible conclusion is narrow: v7 establishes local arithmetic and median timing improvements without a substantial measured tracked allocation reduction. It neither proves a smaller asymptotic space bound nor proves that the full source algorithm's space bound is preserved by every possible implementation of these reductions.

10 Failure cases and boundaries of validity

The following distinctions are important when interpreting the success tables.

1. **Performance is not uniformly improved.** Folding alone has a slower median on the weighted source-precision fixture. A combined P01/DG repeat is also slower than its paired baseline. The sample is too small and variable for universal performance conclusions.

2. **Memory need not decrease.** Two combined source-schedule peaks increase, and all measured changes are below one percent. A smaller algebraic basis is not itself a peak-space theorem.

3. **The quotient requires Boolean observations.** The non-Boolean example fails explicitly. Generic higher-dimensional packets need more than four Boolean coordinates.

4. **The coefficient two cannot be silently removed.** The modulo-eight square example detects that error. Residue-field characteristic must not be substituted for working-ring arithmetic.

5. **Closed summation is not higher-mode truncation.** Dropping the positive modes gives 182305 instead of 424705 in the worked example. The closed expression retains their entire modular effect.

6. **Precision guards remain necessary.** Reducing the numerator too early before division by 16 loses a nonzero result. Closing the geometric loop does not repair that information loss.

7. **The column basis and symmetry certificate must match.** Pairing folded row coefficients with old column monomials, or using an uncertified internal permutation, would be mathematically unjustified. These are interface restrictions, not claims that such incorrect variants were included as successful benchmarks.

8. **Two-position support is special.** The identity $j^b = j$ for $j \in \{0, 1\}$ makes all positive moments coincide. Three or more coordinate values generally produce different moment functions; applying the same one-coordinate coalescence would require a new exact certificate.

9. **The tests are deliberately small.** Direct enumeration is thousands of times faster than the recursive variants on these 16-assignment roots. The suite checks internal mechanisms, not practical superiority on hard SAT instances.

10. **Validation has a defined scope.** The 398 rejected mutations are a useful record/domain audit, not a formal proof of all possible code paths, cache traces, or the complete ETH theorem.

11 Role of Ram Calculus and comparison with conventional methods

In this line of work, the Ram Calculus viewpoint emphasizes replacing a redundant observation representation by an exact quotient, retaining the correct coefficient algebra, and attaching explicit recovery and precision certificates. V7 provides a useful instance of that approach: the representations are reduced without changing the mathematical observable, and the corresponding implementation gains are measured.

However, the specific algebra is also conventional. The four-state step is multilinear reduction modulo $x^2 - x$ and $y^2 - y$, or equivalently Boolean interpolation/Mobius inversion. The closed coefficient is a finite geometric identity in a ring with a nilpotent ratio. Protected division is valuation-aware modular arithmetic, and the packet symmetry step is a certified orbit sum. An otherwise identical conventional implementation can use exactly the same identities and obtain the same operation counts.

Advantages established here. The transformations are explicit, division-free where required, and valid over rings with zero divisors. They preserve local kernel values, not just an aggregate answer. They reduce redundant observation work and remove an avoidable precision-dependent loop. The experiments connect these algebraic statements to exact agreement and measured runtime improvements, with costs of folding and planning included.

Costs and disadvantages. The reductions need domain and symmetry certificates. Basis changes must propagate through both sides of the contraction. Six coefficients are still built before folding. Cache reuse and peak storage are not controlled by basis size alone. The support assumptions are restrictive, and interpolation remains unchanged. Additional implementation and validation complexity is therefore traded for a benefit that is workload dependent.

What is not an established contribution. Neither an accuracy advantage over an exact conventional method nor a Ram-exclusive theorem or computational advantage follows. The counts were already exact. V7 makes their computation cheaper on specified fixtures; it does not make the answers more accurate, prove novelty over all prior literature, or establish that the original manuscript has a gap closed by these experiments.

12 What remains necessary for the full algorithm

12.1 From a packet identity to a uniform algorithmic result

The proofs in this summary are unconditional identities on their stated finite domains. A theorem about the full source algorithm needs further arguments that are not supplied by these packet tests:

1. **Uniform applicability.** Describe the actual packets and coordinate supports appearing throughout the source reduction. Prove which observations admit quotient compression and how their bases can be found and certified at acceptable cost, including general local/crossing mixtures.

2. **Recursive closure.** Show that every generated child retains the required tensor, support, creation-lift, and precision properties. Demonstrate the combined method on connected multi-edge instances with several active recursive levels, not only separate packet and singleton families.

3. **Complete precision accounting.** Carry target recovery, all nonunit divisions, extension-ring representation, creation-time lifts, and child precisions through the full recurrence. V7 preserves the existing packet schedule; it does not prove a new general schedule.

4. **Worst-case symmetry accounting.** Bound certificate discovery and orbit enumeration costs, and include instances with no useful symmetry. P02–P04 already show that packet

interchange cannot be assumed universally.

5. **Bit-complexity accounting.** Count operations at their actual bit sizes, including unit inversions, ring products, planning, certificate construction, child contractions, and interpolation. A local reduction in ring-operation count does not by itself change the leading exponent or asymptotic complexity of the entire algorithm.

6. **Space accounting.** Specify live data, cache limits, streaming/recomputation policy, and native as well as Python storage. Prove a full space recurrence rather than inferring it from small allocation profiles.

7. **End-to-end implementation and evaluation.** Integrate the transforms into the actual SAT reduction and recovery path, compare matched larger instances, and independently audit correctness and resource claims for that implementation.

No improvement to the full ETH algorithm's time exponent, no general SAT upper bound, and no lower bound preventing further improvement are established by v7.

12.2 Concrete future experiments and proposals

The most direct next implementation is to construct the four Boolean coefficients without constructing six intermediate coefficients. This could remove folding overhead, but it has not been measured here and would need the same pointwise and polynomial checks.

A second direction is a certified moment-basis construction for packets with three or more distinct coordinate positions. The relevant question is the exact module of observation functions over the working ring, not numerical rank over a convenient field. A smaller basis must be proved valid for the particular support and precision; generic Boolean support can still require exponentially many coordinates.

Further directions are combined deeper multi-port recursion, connected networks with less favorable cache reuse, and selective-channel Hermite recovery that computes only information needed by the target. The last would require a new recovery theorem: all 116 evaluations remain in v7. Allocation-by-stage measurements, streaming variants, and a more stable timing environment would help distinguish arithmetic savings from storage and host effects. These are proposed extensions, not reported performance results.

13 Evidence provenance and conclusion

This summary was prepared from the v7 manuscript and its saved JSON evidence. Numerical tables were checked against the case–variant records and raw worker measurements; no benchmark was rerun. The evidence set is stored under

`eth-benchmark-v7/results-20261027/`

in the same parent directory as the report. Its principal files are `results.json`, `workers.jsonl`, `instances-and-oracles.json`, `controls.json`, and `validation-audit.json`. The implementation is in `reductions_v7.py`, with the runner and independent validation audit in the same benchmark directory. The v7 delivery also records hashes confirming preservation of the preceding files and retained v6 scientific body.

The central result is precise and useful: two exact representation changes make the selected finite packet routine faster while preserving complete exact outputs. The four-state quotient removes redundant Boolean coordinates; the closed sum removes repeated construction of a coefficient whose higher-mode information is already known algebraically. The combined median gains are 5.24–6.46 under source precision and 1.52–2.07 after the earlier degree-aware and symmetry reductions. There is no substantial measured memory saving, no new precision saving, and no new deeper-recursion result.

Accordingly, v7 is a successful local implementation and validation study, with explicit failure controls and meaningful limitations. Its next scientific step is to prove and test the uniform interfaces, recurrences, and resource bounds required for the complete algorithm, not to extrapolate the small-fixture timing ratios into a general ETH claim.

References

[1] M. Subbarao, with assistance from AI models. *Ram Calculus and the Exponential Time Hypothesis*, revision 7, October 27, 2026. Local file `Ram-Calculus-Exponential-Time-Hypothesis-v7.tex` and corresponding 89-page PDF. New v7 analysis: physical pages 6–14.

[2] M. Subbarao, with assistance from AI models. *Ram Calculus and the Exponential Time Hypothesis*, revision 6, and *Detailed Summary of the v6 ETH Report*, October 27, 2026. Local files `Ram-Calculus-Exponential-Time-Hypothesis-v6.tex` and `Summary-RamCal-ETH-report-v6.tex`.

[3] V7 matched-instance experimental records, October 27, 2026. `eth-benchmark-v7/results-20261027/results.json`; `eth-benchmark-v7/results-20261027/workers.jsonl`; `eth-benchmark-v7/results-20261027/instances-and-oracles.json`.

[4] V7 controls and independent post-audit, October 27, 2026. `eth-benchmark-v7/results-20261027/controls.json`; `eth-benchmark-v7/results-20261027/validation-audit.json`.

[5] V7 implementation and benchmark runner. `eth-benchmark-v7/reductions_v7.py`; `eth-benchmark-v7/run_v7.py`; `eth-benchmark-v7/audit_validation_v7.py`.